GROUP 2B • DATA SCIENCE & ANALYTICS PROJECT

FROM RAW VISIT DATA TO A HOSPITAL OPERATIONS DASHBOARD

The complete project story: problem, data, quality checks, tools, analysis, dashboard and recommendations

TOTAL PATIENTS STUDIED	AVERAGE PATIENT WAIT	ACTIVE DOCTORS	TOTAL CLINIC REVENUE
25,000 Visits	105.90 Minutes (~1 hr 46 min)	40 Physicians	GH₵ 755,945.00

Meet the Team Behind the Analysis

Fourteen interns worked together to take the clinic data from raw database tables to the dashboard and findings in this deck.

01
LA
Louis Adu
TEAM MEMBER

02
MM
Michael Douglas Martey
TEAM MEMBER

03
MS
Michael Shebrah
TEAM MEMBER

04
OP
Oliver Amoaning Pekoh
TEAM MEMBER

05
EA
Emmanuel Akpabli
TEAM MEMBER

06
PA
Precious Bobie Ansah
TEAM MEMBER

07
RP
Robert Vincent Payne
★ GROUP LEADER

08
RK
Regina Kugblenu
TEAM MEMBER

09
NA
Nana Kakra Appiawaa Asamoah
TEAM MEMBER

10
EA
Ernestina Owusu-Ansah
TEAM MEMBER

11
RM
Rita Opokua Mensah
TEAM MEMBER

12
MH
Muhtar Mohammed Habibullah
TEAM MEMBER

13
YM
Yousif Mohammed
TEAM MEMBER

14
LA
Louis Buertey Afotey
TEAM MEMBER

Project Team

• WHY THIS PROJECT

How the project started

The project began with an operational concern, not with a chart.

THE CLINIC'S PROBLEM

Patients were complaining about waiting. Staff had different opinions about which departments were worst.

Visit records existed, but had not yet been analysed to provide an objective answer.

WHY OUR SERVICES WERE NEEDED

- Replace conflicting opinions with evidence.
- Identify departments and visit types associated with longer waits.
- Understand how waiting time changes over time.
- Give administrators a simple monitoring tool.

The deliverable was not just a dashboard—it was a clearer basis for operational decisions.

• PROJECT SCOPE

The question that guided every decision

We kept the work focused on one operational question.

Which departments have the longest patient wait times, and how does that vary by visit type and over time?

1 **How many visits?**

Measure activity and establish scale.

2 **Typical wait?**

Average and median waiting time.

3 **Longest department?**

Compare department averages.

4 **Longest visit type?**

Compare emergency, follow-up, outpatient and unknown.

5 **Trend improving?**

Review time in chronological order.

• DATA SOURCE

Where the data came from

A hosted PostgreSQL training database supplied with the Group 2B clinic-visits brief.

SOURCE SYSTEM

PostgreSQL • internship database

Four source tables represented different parts of clinic operations:

visits	**25,100**	One row per visit: date, patient, doctor, department, visit type, wait minutes, fee.
patients	**900**	Birth year, sex, region and insurance scheme.
doctors	**40**	Doctor name, department and years of experience.
departments	**8**	Department name and floor.

The brief specified read-only access to raw clinic and a separate group2b schema for project outputs.

• DATA QUALITY

The data was not ready for analysis

Loading successfully did not mean the data was trustworthy.

Missing values

Some fields were empty or unavailable.

Duplicates

Repeated records needed investigation.

Inconsistent labels

Categories could be written in different ways.

Mixed date formats

Dates needed consistent parsing.

Impossible values

Some values could not be true or reasonable.

Orphan keys

Records could reference missing dimension records.

Left unchecked, these issues can distort counts, averages, trends and department comparisons.

• THE WORKFLOW

Step 1 — Profile before cleaning

SQL helped us understand the data's size, structure and risks.

01	**Count rows**	Compare table counts with expected starting counts.
02	**Inspect columns**	Check types, keys, nulls and category values.
03	**Check duplicates**	Investigate repeated records.
04	**Check dates and ranges**	Find dates that fail parsing and impossible wait values.
05	**Check relationships**	Look for visits without matching patients, doctors or departments.

Tool: SQL queries in PostgreSQL, accessed through DBeaver.

• THE WORKFLOW

Step 2 — Clean and standardise the data

Cleaning decisions should follow the issues found during profiling.

DATES
Parse mixed date formats consistently.

CATEGORIES
Standardise spelling and labels, including visit types.

DUPLICATES
Investigate repeated rows and document the decision.

VALIDITY
Review impossible values; correct, exclude or flag based on documented rules.

KEYS
Check patient, doctor and department references.

TRACEABILITY
Record decisions so changes can be explained.

Tool: Python cleaning notebook; cleaned outputs were written to the group's schema.

• QUALITY CONTROL

A quality checkpoint — reconcile the counts

The source brief and final dashboard show different visit totals.

EXPECTED SOURCE VISITS

25,100

Starting count in the project brief.

FINAL DASHBOARD VISITS

25,000

Count shown in the final Power BI view.

DIFFERENCE

100

Must be traceable to documented cleaning decisions.

What We Found in the Raw Data, and How We Fixed It

FINDING 1

Duplicate Records

Exact duplicate visit rows were inflating patient counts and revenue.

Removed 100 duplicates: 25,100 → 25,000 visits

FINDING 2

Mixed Date Formats

Visit dates were stored as text in four different formats, which breaks sorting and monthly trends.

Converted to one date format (day-first)

FINDING 3

Impossible Wait Times

Some records showed negative waiting times, which cannot happen and would distort the 105.9-minute average

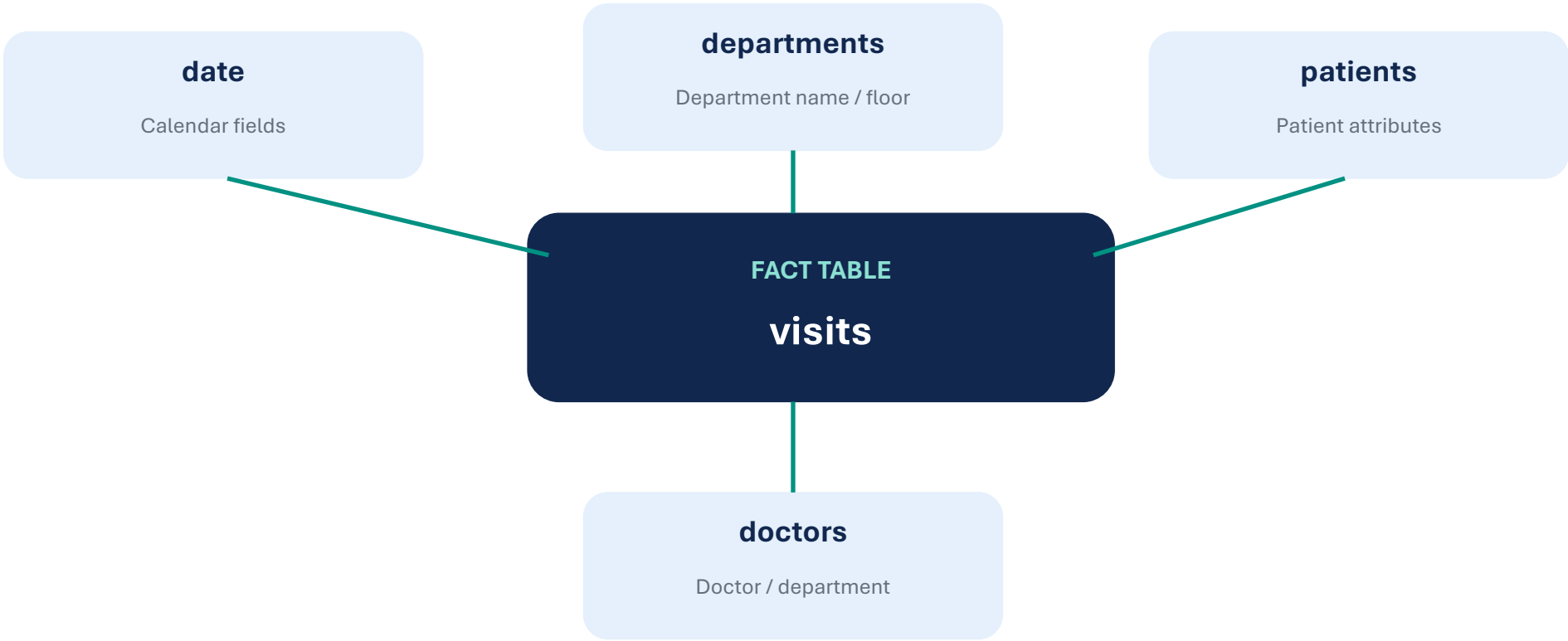
Reviewed individually and handled explicitly

Also checked: inconsistent text (capitalisation and stray spaces in categories), patient IDs with no matching patient record, and missing values.

• THE WORKFLOW

Step 3 — Build a model that keeps numbers consistent

We connected visit records to descriptive tables instead of relying on one flat table.



The date table enables reliable calendar filtering and month-by-month comparisons.

• THE WORKFLOW

Step 4 — Build, test, improve and simplify

The first dashboard was a draft; the final version was shaped by iteration.

WEEK 1	WEEK 2	WEEK 3	WEEK 4	WEEK 5	WEEK 6	WEEK 7–8
Scope	**Profile**	**Clean**	**Model**	**Draft**	**Iterate**	**Present**
Agree the question.	Find data issues.	Standardise data.	Build relationships.	Create measures/visuals.	Fix and improve readability.	Freeze, polish and rehearse.

The Clinic at a Glance: Key Findings

Over two years (2024 to 2025), our clinic cared for 25,000 patients across 8 departments. Here are the facts everyone needs to know:

TOTAL VISITS

25,000

Consistently steady demand of ~1,000 patients every single month.

AVERAGE WAIT

105.90 min

TYPICAL WAIT TIME

About 1 hour 46 minutes.

Delays are everyday, not just on bad days.

MEDIAN WAIT

106 min

Middle wait value.

EMERGENCY RATE

14.45%

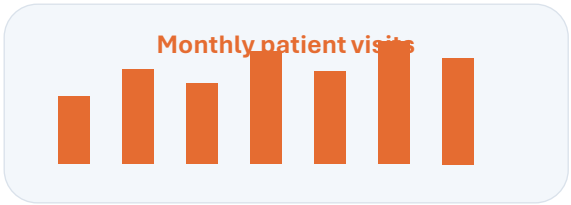
3,613 emergency cases arrived at our facility during this period

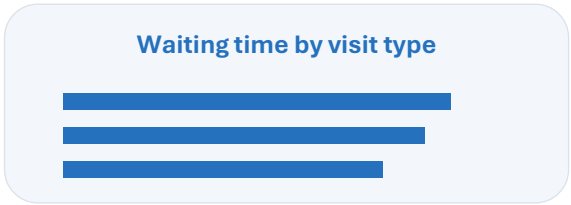
FEE REVENUE

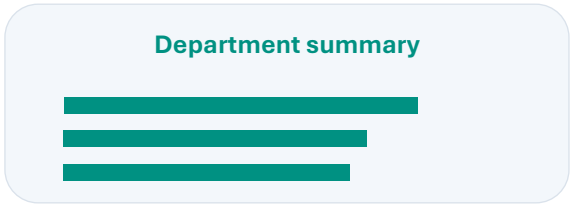
GH₵755,945

Strong and consistent revenue generation across all 8 departments

MONTHLY TRENDS • DEPARTMENT SUMMARY • WAIT BY VISIT TYPE





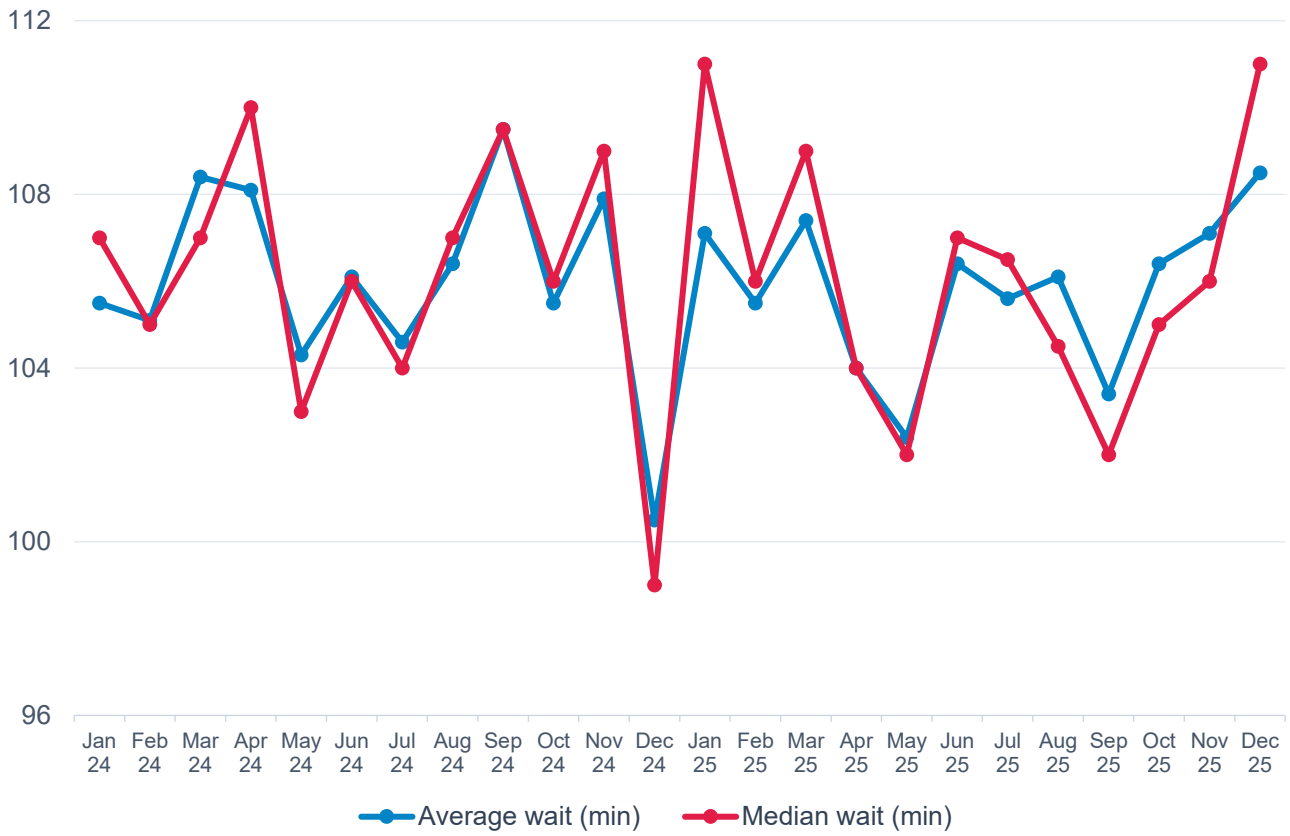


The actual Power BI page is interactive; figures can change when filters are applied.

OVER TIME: MONTHLY WAITING TIMES

Waiting Times Over Time: No Trend, No Season

The project question also asked how waits vary over time. Month by month, the average wait stays between about 100 and 110 minutes across the two years.



Axis zoomed to 96–112 minutes so the month-to-month movement is visible.

LOWEST MONTH

Dec 2024: 100.5 min

HIGHEST MONTH

Sep 2024: 109.5 min

VISITS PER MONTH

About 1,000 to 1,100, steady

Waits are not getting better or worse over time, and there is no seasonal peak. A statistical test across months finds no meaningful change (p = 0.20).

ISSUE #1: WAITING TIMES

Patients Wait Nearly 1 Hour and 46 Minutes Everywhere

We analyzed waiting times by department and expected big differences. Instead, we found that every department has the same bottleneck.

Average Wait Time by Department (in Minutes)

Department	Total Visits	Avg Wait Time	Status
Laboratory	3,105	**107.26 min**	**Longest**
Emergency	3,153	**106.01 min**	**High**
Paediatrics	3,061	**106.14 min**	**High**
Ophthalmology	3,097	**105.86 min**	**High**
Maternity	3,179	**105.73 min**	**High**
Physiotherapy	3,132	**105.49 min**	**High**
Dental	3,129	**105.39 min**	**High**
General Medicine	3,144	**105.34 min**	**High**
Clinic Total	**25,000**	**105.90 min**	**All Elevated**

What This Means in Plain English

Whether a patient is getting a simple tooth cleaning in Dental (105.4m) or having severe pain in Emergency (106.0m), they wait about the same hour and three-quarters.

Laboratory is the Slowest Point

Laboratory has the highest wait time at 107.26 minutes (median 108.0m). Many patients from other departments wait here for tests, compounding total clinic delays.

Even Load Distribution

All 8 departments handle approximately 3,100 visits each (12.2% to 12.7% of all patients). No single department is being neglected or overburdened.

Departmental Waiting Times

ISSUE #2: URGENT CARE FINDINGS

The Emergency Paradox: Urgent Cases Wait Just as Long

Common sense says emergency patients should be seen much faster than routine visits. Our clinic data reveals an urgent priority problem.

OUTPATIENT VISITS

105.53 min

Routine consultations and initial walk-in checkups.

FOLLOW-UP VISITS

105.99 min

Scheduled follow-ups and returning patients.

EMERGENCY VISITS

106.59 min

Urgent, acute conditions needing immediate attention.

Why is this happening?

Emergency visits make up 14.45% of our entire clinic volume (over 3,600 patients). Right now, emergency cases join the exact same administrative registration and reception queue as routine check-ups. Because there is no separate triage desk, emergency cases get stuck behind non-urgent administrative processing.

Visit Type Analysis

Are Doctors Overworked? Myth vs. Fact

When waiting times are high, people often assume: "We need more doctors" or "Certain doctors are overloaded." The data proves this is false.

The Facts From Our Data

- We have 40 active doctors across the clinic.
- Every doctor sees an average of 625 patients over the 2-year period.
- Doctor patient loads are remarkably equal:
  - Highest: Dr. Isaac Asante (644 visits)
  - Lowest: Dr. Joyce Oppong (586 visits)
- All 40 doctors rotate across all departments, seeing between 77 and 79 patients per department.

Workload Does Not Drive Wait Times

If doctors were the bottleneck, departments with more patients per doctor would have higher waiting times. Let's compare:

Maternity: Highest load (79 visits/doctor) → 105.7 min wait

Paediatrics: Lowest load (77 visits/doctor) → 106.1 min wait

Verdict: Having fewer patients per doctor in Paediatrics did not make wait times any faster. The delay occurs **before** patients enter the doctor's room.

Doctor Workload vs. Wait Times

OPERATIONAL DIAGNOSIS & HYPOTHESIS

Where Is the Time Lost? Finding the Bottleneck

Our data shows doctors aren't the problem. Since every patient waits roughly the same 106 minutes, the delay must happen during shared steps before seeing a doctor:

CHECKPOINT 1

Front Desk & Check-In

Manual patient lookup, registration queues, and paperwork during peak morning hours.

+

CHECKPOINT 2

Nursing & Vitals Desk

All patients—emergency and routine—wait in the exact same line to get their vitals taken.

+

CHECKPOINT 3

Lab & Test Waiting

Lab has the longest delay (107 min), meaning test queues add extra waiting time.

=

ALREADY PROVEN

Doctor Consultation

Our 40 doctors handle an even, manageable workload. The delay happens before the exam room.

In Summary: When a delay happens equally in every department, it points to common steps every patient goes through on arrival. Our next move is a simple 2-week check-in audit to measure where the minutes are actually lost.

Investigative Flow Analysis

ISSUE #4: DATA INTEGRITY

Data Quality Audit: The 6% Missing Category Problem

Our data review discovered that 1,500 patient records were recorded without specifying whether they were Outpatient, Follow-Up, or Emergency.

Missing Visit Types by Department

Department	Total Visits	Missing Records	Missing %
Laboratory	3,105	202	**6.51%**
Paediatrics	3,061	194	**6.34%**
Maternity	3,179	196	**6.17%**
Emergency	3,153	194	**6.15%**
Ophthalmology	3,097	188	**6.07%**
Physiotherapy	3,132	180	**5.75%**
Dental	3,129	178	**5.69%**
General Medicine	3,144	168	**5.34%**
Total Clinic	**25,000**	**1,500**	**6.00%**

Data Quality Review

1. It's a Software Issue, Not Staff Laziness

The error rate is almost identical across all departments (~5.3% to 6.5%) and occurs every single month (ranging from 4.4% in Aug 2025 to 8.0% in Oct 2024). This proves our system allows staff to save records without selecting a visit type.

2. How We Protected the Analysis

These 1,500 visits were fully counted for revenue (GH₵ 755.9k) and department visit volumes, but excluded only when comparing visit types (Outpatient vs. Follow-up vs. Emergency) so the percentages remain 100% accurate.

3. The Easy Fix

Make "Visit Type" a mandatory required field in the software so records cannot be saved without it.

NEXT STEPS

4 Simple Steps to Cut Down Wait Times

Clear, practical actions to test our findings and start getting patients through faster.

1. Fast-Track Emergency Patients

Create a separate check-in for urgent cases right away. Emergency patients should never have to wait in the regular checkup line.

2. Track Where the Time Goes

Run a quick 2-week check on reception, vitals, and lab queues to see exactly how many minutes are lost at each desk before seeing the doctor.

3. Fix the Computer Form

Make "Visit Type" a required field in the system so staff cannot save a record with a blank category. This solves our missing 6% data immediately.

4. Speed Up the Lab Queue

The Lab has our highest wait time (107 min). Reviewing how samples are taken and results sent back will help free up the whole clinic.

Our Goal: Clear the front-desk bottlenecks and bring overall wait times under 45 minutes.

Strategic Next Steps

• TOOLS & DELIVERABLES

Tools used and what each one contributed

Each tool had a defined role in the end-to-end workflow.

PostgreSQL	Hosted database containing raw tables and cleaned project outputs.
DBeaver	Connected to PostgreSQL and ran SQL profiling/validation queries.
Python notebook	Cleaned and standardised data based on profiling findings.
Power BI Desktop	Built the date table, relationships, DAX measures, visuals and slicers.
Git / GitHub	Versioned work and stored SQL, notebook, notes, README and dashboard.

Outputs: cleaned analytical data, quality notes, a working model and an administrator-friendly dashboard.

THANK YOU

Questions & Discussion

We used data to move from assumptions about waiting times toward evidence and practical next steps.

From data quality to decisions—one step at a time.